

Wideband CDMA - W-CDMA

The CDMA in W-CDMA refers to the multiplexing technique, and not Qualcomm's cdmaOne standard. The W-CDMA standard uses CDMA to achieve the 144 kbps - 2 Mbps data rates that define a 3G network.

We visited the concept of a full-duplex channel at the beginning of this chapter - it allowed users to transmit and receive data simultaneously. There are two ways of 'duplexing' a channel - *Time Division Duplexing (TDD)*, which uses TDMA to separate the incoming and outgoing data, and *Frequency Division Duplexing (FDD)*, which uses FDMA to separate them. CDMA networks thus far had used TDD in 1.25 MHz of bandwidth. W-CDMA, however, uses FDD - two 5 MHz frequency bands to achieve much higher capacity and speeds for data transmission.

The first network to employ the W-CDMA technique was FOMA - Freedom of Mobile Multimedia Access, developed by Japanese company NTT DoCoMo.

It must be noted that W-CDMA is not compatible with any of Qualcomm's CDMA standards.

The Universal Mobile Telephone System - UMTS

UMTS is basically W-CDMA deployed on existing GSM infrastructure. Adding to the security of the CDMA technique itself is the USIM - Universal Subscriber Identity Module, which is a more secure version of today's SIM cards, and with more memory.

UMTS networks will have the 'soft hand-off' we saw only in the cdmaOne networks so far. Hand-offs will also be possible between UMTS and other 3G technologies, between FDD and TDD systems and between UMTS and GSM.

Such advances, however, come at the cost of a very challenging and very expensive implementation. We can only wait with bated breath. Having now looked at what will take GSM networks into the third generation, we now move on to cdmaOne networks.